AquaRun

Community Water Refill & Delivery System · Technical Summary

Caloocan City · October 2026
System Specification

1. SYSTEM SUMMARY

AquaRun is a web-based water refill ordering, dispatch, and bottle tracking system designed for local refilling stations. It replaces paper receipts and verbal phone orders with an automated workflow: customers book refills online, station staff manage purification and batch dispatch, riders deliver water with live mobile tracking, and empty gallon swaps and cash collections are reconciled automatically.

2. TECH STACK & APIS USED

Area	Technology	Purpose
Frontend	Vanilla JavaScript (ES Modules), HTML5, CSS3	Lightweight, fast client without heavy frameworks. Direct modular scripts (`js/core` , `js/pages` , `js/services`).
Backend & DB	Supabase (PostgreSQL 15+)	Relational database, table constraints, triggers, and Row Level Security (RLS) policies.
Authentication	Supabase Auth (JWT)	Secure customer, staff, rider, and admin logins with role-based routing.
Realtime Sync	Supabase Realtime + BroadcastChannel	WebSockets for live order updates + browser BroadcastChannel for instant cross-tab sync.
APIs / Stored Procs	PostgreSQL RPC Functions	Atomic transactional business logic (`create_order` , `mark_ready` , `record_delivery` , `staff_approve_remittance`).
Hosting	Vercel / Netlify Edge	Continuous deployment from GitHub with custom cache-control headers.

3. USER ROLES & RESPONSIBILITIES

Customer
Browses products, selects bottle swap or new container, picks delivery slot, and tracks order status.

Station Staff
Validates orders, manages refill queue, assigns batches to riders, and verifies cash remittances.

Rider
Follows delivery run sheet, delivers gallons, logs returned empty bottles, and submits cash collected.

Admin
Monitors overall sales, hourly demand patterns, slot capacities, product prices, and user accounts.

4. KEY BUSINESS LOGIC

1:1 Empty Bottle Swap Ledger
Tracks gallons delivered vs. empties collected per customer. Net held bottles update automatically to prevent bottle loss.

Dual Container Pricing Engine
Refill price (₱30–₱35) if customer returns an empty gallon; full price (refill + ₱150–₱220 container fee) if purchasing new container.

5. END-TO-END SYSTEM FLOW

The diagram below illustrates the exact step-by-step lifecycle of an order—from customer checkout to station processing, batch rider assignment, doorstep delivery, and cash remittance closeout:

CUSTOMER STEP 1	**Places Order Online**	STATUS: SUBMITTED
	Customer selects water product (Round Gallon, Slim Gallon, or Bottle), specifies whether they are swapping an empty gallon or buying a new container, enters delivery address, and chooses a morning (8–12 AM) or afternoon (1–5 PM) time slot.	
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STATION STAFF STEP 2	**Confirms Order & Validates Slot**	STATUS: CONFIRMED
	Order appears in real time on the Staff Order Queue. Staff verifies slot capacity and delivery address. Staff confirms the order (or adjusts the time slot if needed).	
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STATION STAFF STEP 3	**Washes, Refills & Packages Gallons**	STATUS: REFILLING → READY
	Station personnel sanitize the containers, refill them using multi-stage reverse osmosis and UV purification, seal the caps, and click "Mark Ready" on the workbench.	
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STATION STAFF STEP 4	**Smart Batch Dispatch to Rider**	STATUS: RIDER ASSIGNED → OUT FOR DELIVERY
	Staff groups ready orders in the same barangay or delivery corridor into a batch run and assigns them to an available rider. The rider receives the delivery stops instantly on their mobile phone.	
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RIDER STEP 5	**Doorstep Delivery & Empty Bottle Collection**	STATUS: DELIVERED
	Rider arrives at the customer's home, hands over purified gallons, and collects empty swap bottles. On the mobile runsheet, rider logs exact bottles delivered and empties collected (updating the customer's bottle ledger).	
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RIDER STEP 6	**Cash Collection & Remittance Submission**	PAYMENT: CASH ON DELIVERY (COD)
	Rider collects cash payment from the customer. At the end of the delivery run, the rider returns to the station and submits their cash turnover report via the mobile remittance screen.	
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STATION STAFF STEP 7	**Cash Verification & Final Settlement**	STATUS: COMPLETED & SETTLED
	Station staff physically counts the cash collected by the rider against the digital total in the Cash & Sales workbench. Staff clicks "Verify & Accept Cash", completing the order and settling the shift accounts.	